1A 150KHz 100V Buck DC to DC Converter XL7035

Features

- Operating Voltage: 10V~80V
- Output Adjustable from 1.25V to 20V
- Maximum Duty Cycle 100%
- Minimum Drop Out 1V
- Fixed 150KHz Switching Frequency
- Maximum 1A Output Current
- Recommend Maximum 1A Output Current At 5V Output,48V Input
- Recommend Maximum 0.6A Output Current At 15V Output,48V Input
- Internal Optimize HV Power Transistor
- High efficiency up to 85%
- Excellent line and load regulation
- Built in thermal shutdown function
- Built in output short Protection Function
- Built in current limit function
- TO263-5L package

General Description

The XL7035 is a 150KHz fixed frequency PWM buck (step-down) DC/DC converter, capable of driving a 1A load with high efficiency, low ripple and excellent line and load regulation. Requiring a minimum number of external components, the regulator is simple to use and include internal frequency compensation and a fixed-frequency oscillator.

The PWM control circuit is able to adjust the duty ratio linearly from 0 to 100%. An over current protection function is built inside. When output short protection function happens, the operation frequency will be reduced from 150KHz to 45KHz. An internal compensation block is built in to minimize external component count.

Applications

- Ebike Controller Power Supply
- Telecom / Networking Equipment

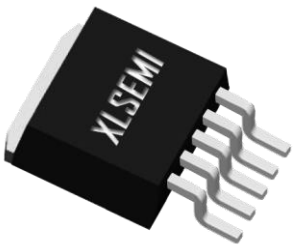


Figure1. Package Type of XL7035

Pin Configurations

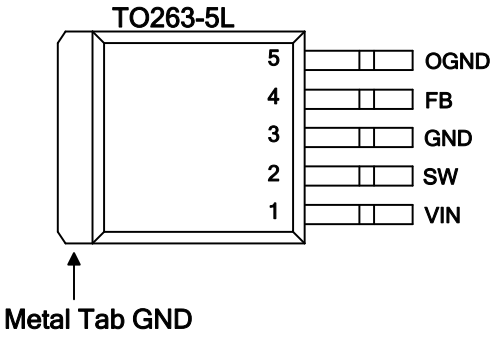


Figure2. Pin Configuration of XL7035 (Top View)

Table 1 Pin Description

Pin Number	Pin Name	Description
1	VIN	Supply Voltage Input Pin. Bypass Vin to GND with a suitably large capacitor to eliminate noise on the input.
2	SW	Power Switch Output Pin (SW). Output is the switch node that supplies power to the output.
3	GND	Ground Pin. Care must be taken in layout. This pin should be placed outside of the Schottky Diode to output capacitor ground path to prevent switching current spikes from inducing voltage noise into XL7035. The exposed PAD is GND.
4	FB	Feedback Pin (FB). Through an external resistor divider network, Feedback senses the output voltage and regulates it. The feedback threshold voltage is 1.25V.
5	OGND	Output Ground Pin.

Function Block

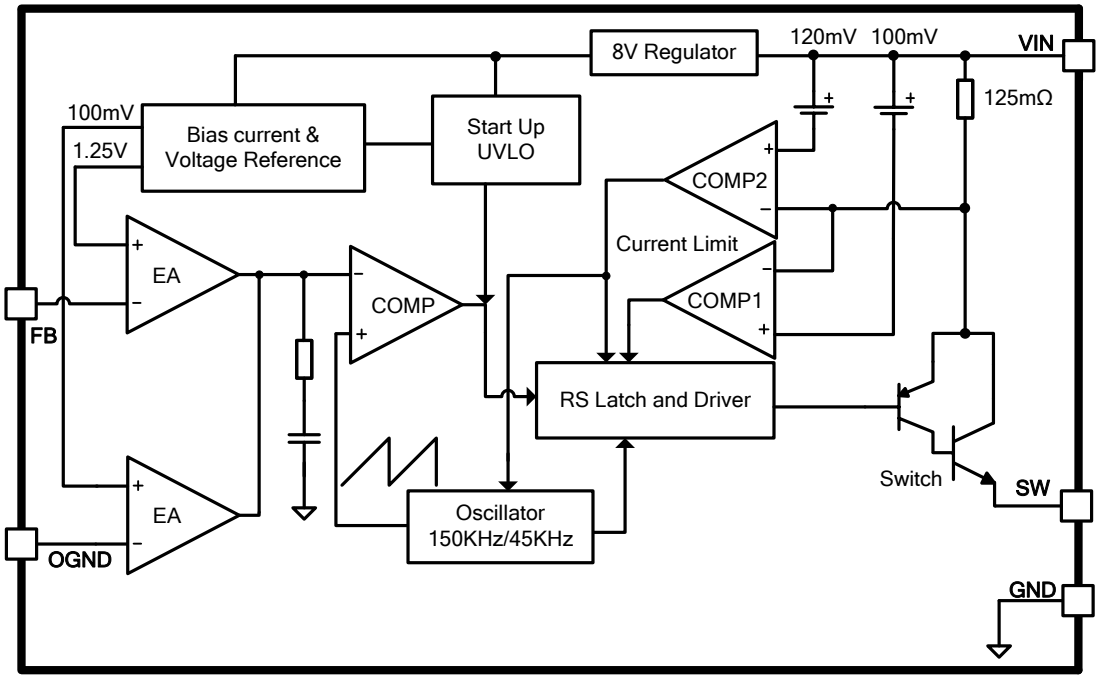
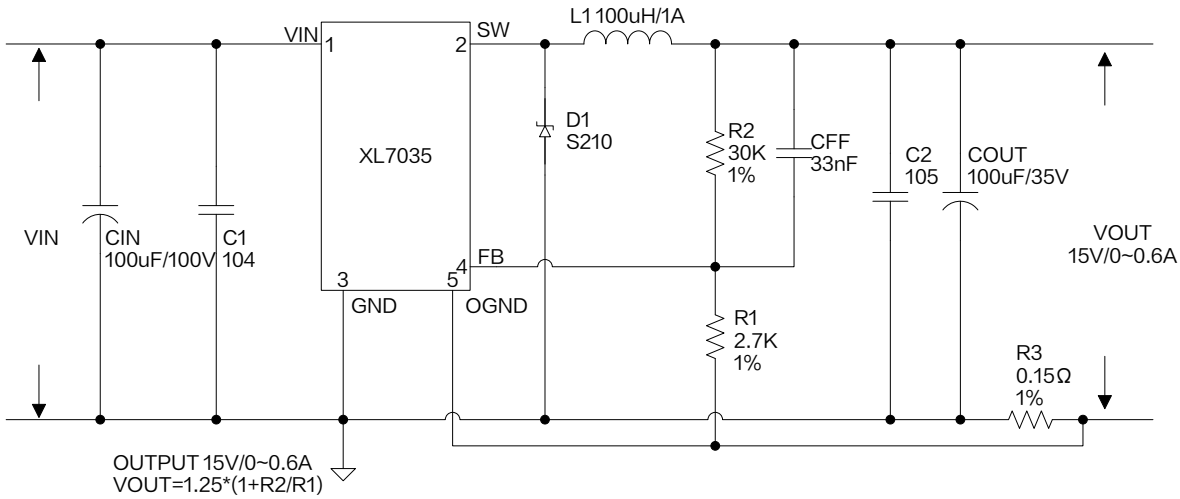


Figure3. Function Block Diagram of XL7035

Typical Application Circuit



R3 for limit maximum output current, R3 choose 91 mohm in VOUT≤5V application; R3 choose 150 mohm in VOUT>5V application.

Figure4. XL7035 Typical Application Circuit

Ordering Information

Order Information	Marking ID	Package Type	Eco Plan	Packing Type Supplied As
XL7035E1	XL7035E1	TO263-5L	RoHS & HF	800 Units on Tape & Reel

Absolute Maximum Ratings（Note1）

Parameter	Symbol	Value	Unit
Input Voltage	Vin	−0.3 to 100	V
Feedback Pin Voltage	V_{FB}	−0.3 to Vin	V
OGND Pin Voltage	V_{OGND}	−0.3 to 7	V
Output Switch Pin Voltage	V_{SW}	−0.3 to Vin	V
Power Dissipation	P_D	Internally limited	mW
Thermal Resistance (TO263-5L) (Junction to Ambient, No Heatsink, Free Air)	R_{JA}	30	°C/W
Maximum Junction Temperature	T_J	−40 to 150	°C
Operating Junction Temperature	T_J	−40 to 125	°C
Storage Temperature	T_{STG}	−65 to 150	°C
Lead Temperature (Soldering, 10 sec)	T_{LEAD}	260	°C

Note1: Stresses greater than those listed under Maximum Ratings may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operation is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

XL7035 Electrical Characteristics

T_a = 25℃;unless otherwise specified.

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
System parameters test circuit figure4						
VFB	FB Voltage	Vin =20V to 80V,Vout=15V Iload=0.1A to 0.6A	1.225	1.25	1.275	V
η	Efficiency	Vin=36V ,Vout=15V Iout=0.6A	–	87	–	%
η	Efficiency	Vin=48V ,Vout=15V Iout=0.6A	–	84	–	%
η	Efficiency	Vin=60V ,Vout=15V Iout=0.6A	–	82	–	%

Electrical Characteristics (DC Parameters)

Vin = 48V, GND=0V, Vin & GND parallel connect a 33uf/100V capacitor; Iout=0.2A, T_a = 25℃; the others floating unless otherwise specified.

Parameters	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Input operation voltage	Vin		10		80	V
VIN UVLO	Vin_uvlo			8	9	V
Quiescent Supply Current	I_q	V_{FB}=2V		2.5	5	mA
Oscillator Frequency	Fosc		120	150	180	KHz
Switch Current Limit	I_L	V_{FB}=0V R3=0.091Ω		1.099		A
Output Saturation Voltage	V_{CE}	V_{FB}=0V I_{SW}=1A		1.5		V
Max. Duty Cycle	D_{MAX}	V_{FB}=0V		100		%

Typical Performance Characteristics

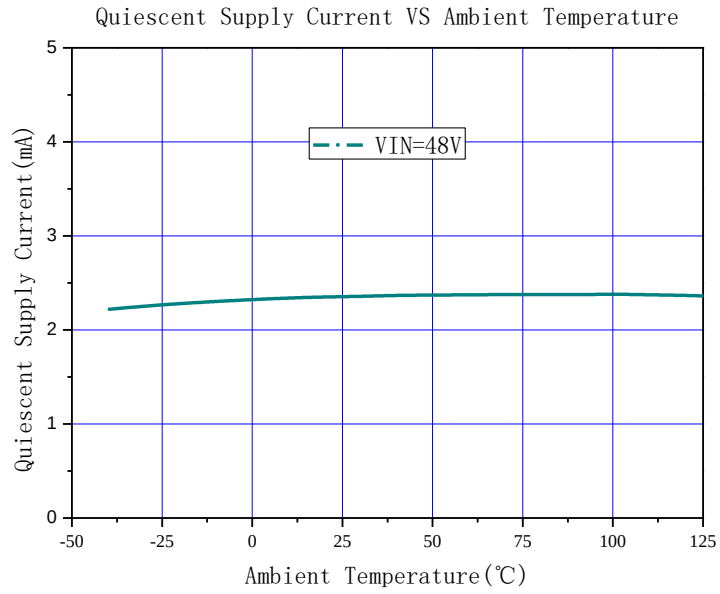


Figure5. Quiescent Current Curve

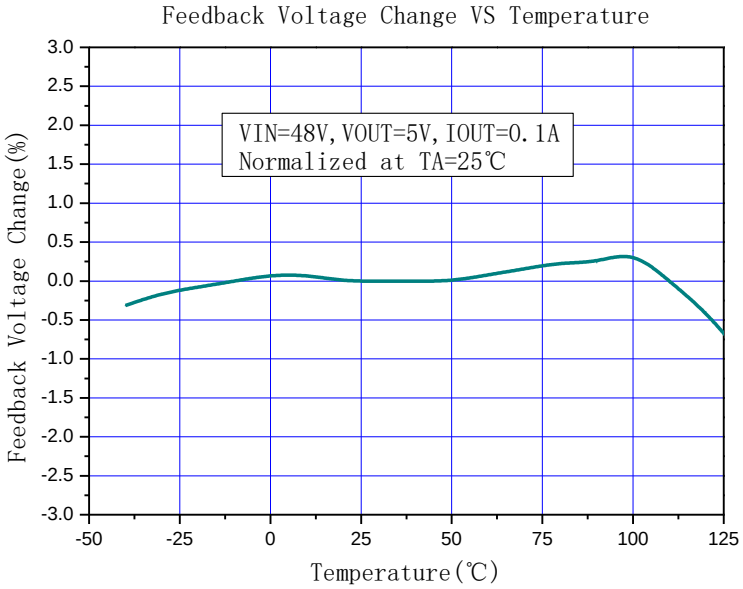


Figure6. Feedback Voltage Curve

Typical System Application(VOUT=15V/0.6A)

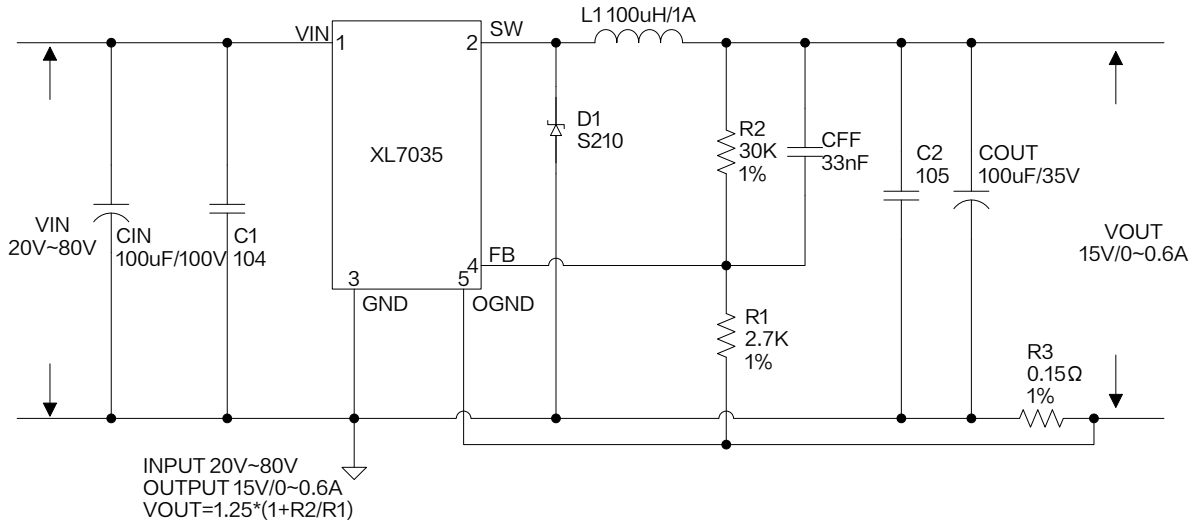


Figure7. XL7035 System Application (VIN=20V~80V, VOUT=15V/0.6A)

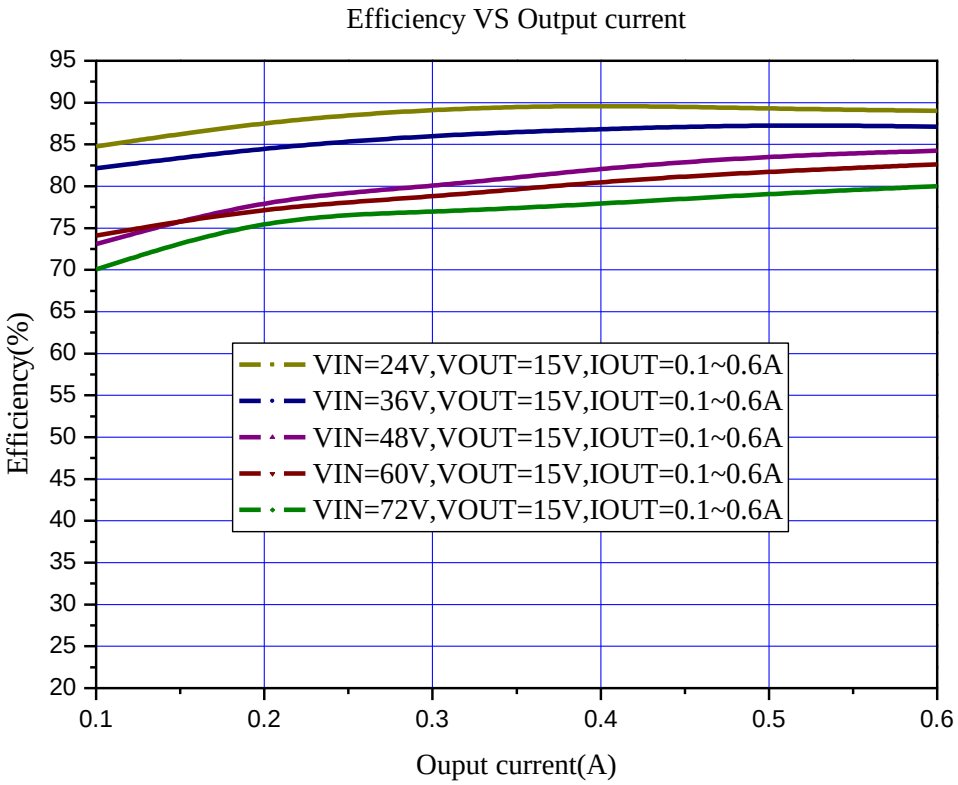


Figure8. XL7035 System Application (Efficiency VS Output Current)

Typical System Application(VOUT=5V/1A)

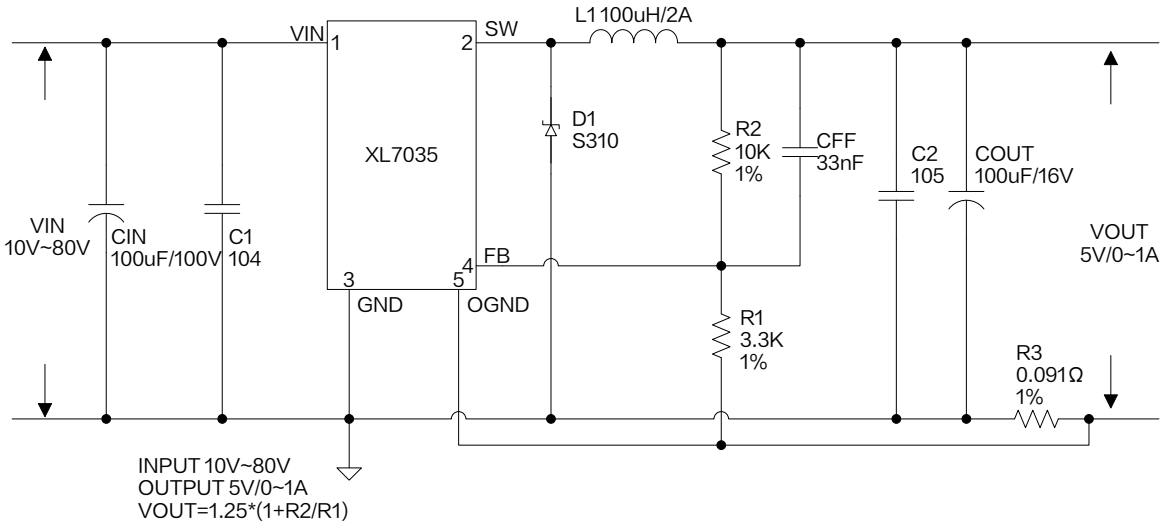


Figure9. XL7035 System Application (VIN=10V~80V, VOUT=5V/1A)

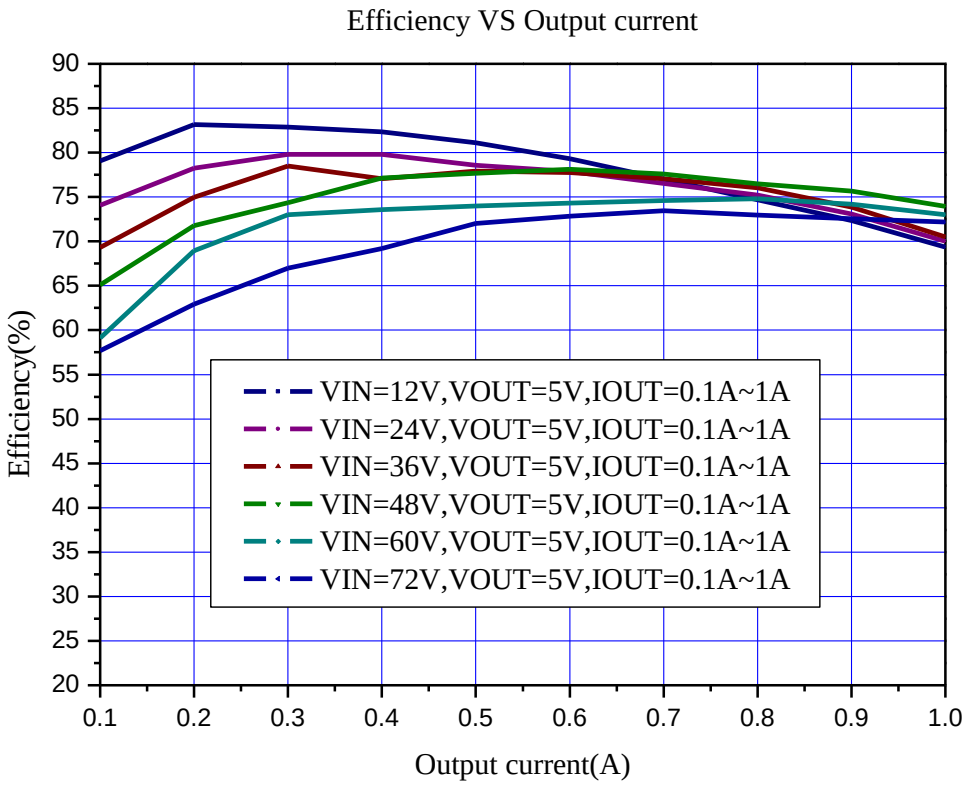
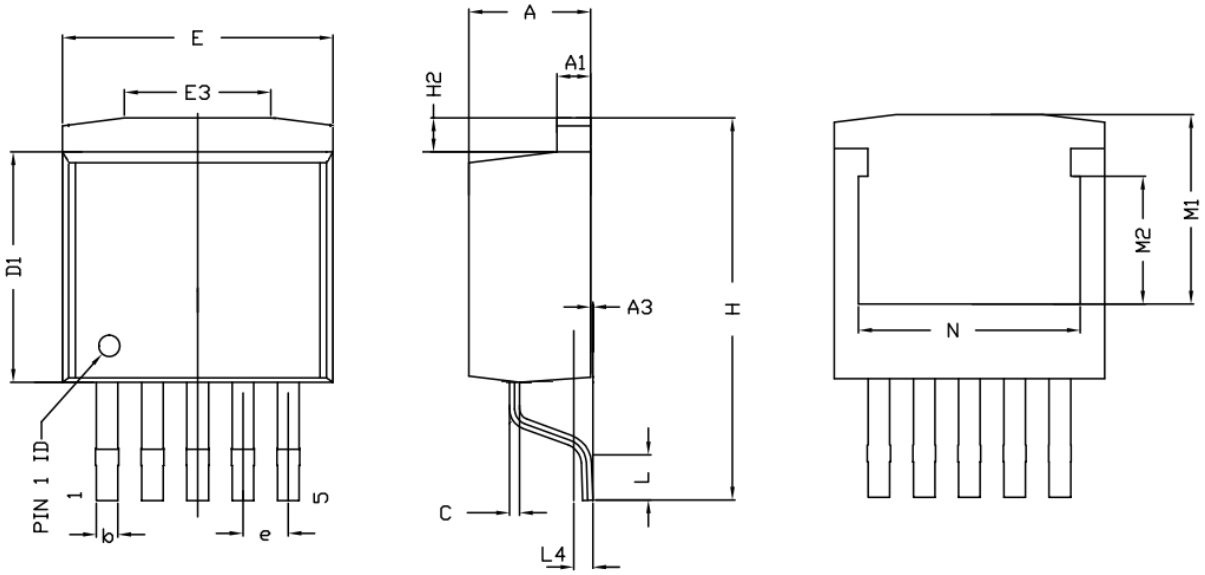


Figure10. XL7035 System Application (Efficiency VS Output Current)

Package Information

TO263-5L



Symbol	Dimensions In Millimeters			Dimensions In Inches		
	Min.	Nom.	Max.	Min.	Nom.	Max.
A	4.37	4.57	4.77	0.172	0.180	0.188
A1	1.17	1.27	1.42	0.046	0.050	0.056
A3	0.00	0.13	0.25	0.000	0.005	0.010
b	0.71	0.81	0.97	0.028	0.032	0.038
c	0.33	0.38	0.76	0.013	0.015	0.030
D1	8.38	8.70	9.00	0.330	0.343	0.354
E	9.90	10.16	10.39	0.390	0.400	0.410
E3	5.00	6.50	8.00	0.197	0.256	0.315
e	1.70 REF.			0.067 REF.		
H	13.00	13.85	14.35	0.511	0.545	0.565
H2	0.90	1.27	1.42	0.035	0.050	0.056
L	1.68	1.98	2.28	0.066	0.078	0.090
L4	0.56	0.76	0.96	0.022	0.030	0.038
M1	6.00	7.11	8.00	0.236	0.280	0.315
M2	–	4.80	–	–	0.189	–
N	7.30	8.33	9.30	0.287	0.328	0.366

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